

COMMON COURSE ASSIGNMENT FORMAT

A. Assignment Information

Particular	Details
Department:	BTECH - AI – ML
Programme:	Bachelor of Technology
Course Code & Course Name	ITA0401 – STATISTICS WITH R PROGRAMMING
Academic Year / Batch	2026-27
Faculty Name	Vignesh Janarthanan
Assignment Title	E-Commerce Order and Customer Analytics System
Date of Issue	02-09-2026
Date of Submission	02-09-2026
Maximum Marks	100
Course Outcome(s) – CO	CO1: Apply R programming fundamentals including data structures, vectors, matrices, arrays, and class types to perform mathematical operations and manage data effectively. Identify and implement appropriate control structures to solve a particular programming problem CO2: Implement looping constructs, control structures, and recursive functions in R to solve computational problems with efficient program logic. Analyse the Correlation between the parameters in a dataset and establish the analysis.(2,3,4) CO3: Construct and manipulate data frames using reshaping, merging, and file I/O operations in R to prepare structured datasets for analysis CO5: Develop data visualizations and regression models in R using plotting libraries and machine learning techniques including decision trees to interpret and present analytical insights.
Bloom's Taxonomy Level	L3 – Apply; L4 – Analyse
SDG Mapping (SDG 1-17)	SDG 7 – Supports efficient e-commerce operations, business growth, and improved productivity through automated order and sales management.; SDG 11 – Promotes digital innovation and technology-driven management of customers, products, inventory, and orders. SDG 12 – Supports responsible consumption and production by monitoring inventory, identifying low-stock products, reducing unnecessary stock, and analyzing purchasing patterns.

Industry / Societal Relevance	E-Commerce Industry Retail Analytics Business Intelligence Inventory Management Customer Relationship Management
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B. Assignment Problem / Challenge

- An online shopping platform needs a reliable R-based system for handling customers, products, inventory and orders while converting transaction records into useful business decisions.
- The challenge is to combine R programming fundamentals, object-oriented design, functions, data-frame operations, file handling and statistical visualisation into one menu-driven application.
- The system should support order operations and then analyse the resulting data to identify revenue trends, best-selling products, customer purchase behaviour and low-stock items.

Challenge Objectives

- Represent Customers, Products and Orders using suitable R data structures and classes.
- Implement registration, product management, inventory updates, order placement, cancellation and payment-related operations.
- Apply category-specific discount and return rules using reusable methods.
- Use data frames, filtering, sorting, merging, reshaping and file I/O to maintain analytical datasets.
- Generate revenue, customer, product and inventory summaries using R visualisation tools.
- Provide input validation and error handling so invalid operations do not corrupt the dataset.
- Produce a menu-driven workflow that is understandable to a non-technical e-commerce operator.

Why This Challenge Requires Multiple R Techniques

- The problem is broader than a single calculation. Data structures manage entities, classes represent domain objects, functions modularise operations, data frames organise records, file handling supports persistence, and visualisation converts records into decisions.
- The workflow also mirrors an e-commerce control room: orders feed revenue calculations, product records support stock monitoring, customer records support purchase analysis, and reports provide a compact view of business performance.

Industry and Societal Relevance

- E-Commerce Industry – order processing, inventory control and sales analytics.
- Retail Analytics – category performance, customer value and product demand analysis.
- Business Intelligence – converting transaction records into operational KPIs.
- Inventory Management – detecting low-stock products before stock-outs occur.
- Customer Relationship Management – comparing purchase frequency and spending.
- Responsible Consumption – using sales and stock information to reduce unnecessary inventory.

C. Problem Statement

- Design, implement and evaluate a menu-driven R application for an online shopping platform containing customers, products and orders.
- The application shall support customer registration, product catalogue management, stock updates, order placement, cancellation, discount calculation, payment recording and report generation.
- Customers, Products and Orders shall be represented using suitable R data structures and object-oriented techniques. Product categories shall support different business rules through reusable methods.
- The application shall analyse completed transactions to calculate revenue, identify best-selling products, compare customer purchases, detect low-stock products and generate graphical reports.
- The system shall use input validation, error handling, functions, data-frame operations and file I/O so that the application remains modular and dependable.

Expected Output

- A working menu-driven R prototype.
- Structured customer, product and order data frames.
- Reusable functions and suitable R classes.
- Transaction summaries and customer/product reports.
- Revenue and sales visualisations.
- Low-stock alerts and best-seller analysis.
- CSV import/export evidence and a reproducible analysis workflow.

D. Requirements and Constraints

Functional Requirements

- Register and search customers.
- Add, update and inspect products.
- Maintain stock quantities after sales and cancellations.
- Place valid orders and record payment mode.
- Cancel eligible orders and reverse inventory where required.
- Apply category-specific discount rules.
- Calculate order totals, completed revenue and customer spending.
- Identify best-selling products and low-stock items.
- Generate graphical and tabular reports.
- Export analytical data to CSV files.

Performance Requirements

- Menu operations should return results without unnecessary repeated calculations.
- Reports should be generated from the current dataset rather than hard-coded values.
- Filtering and aggregation should remain practical for thousands of records if the prototype is scaled.
- Analytical calculations should use consistent definitions for completed and cancelled orders.

Technical Constraints

- The core implementation must be written in R.
- Base R data structures and data frames should be used where appropriate.
- Reusable functions must be used instead of repeating the same business logic.

- The solution should demonstrate an R object-oriented approach such as S3 classes.
- CSV file I/O must be included for persistence and reporting.

Time and Resource Constraints

- The prototype should run on a standard student laptop.
- The dataset should be manageable for demonstration while the logic should be scalable.
- The implementation should prioritise correctness and clear output over unnecessary interface complexity.

Reliability and Error Handling

- Reject invalid customer IDs, product IDs and non-positive quantities.
- Prevent orders when available stock is insufficient.
- Prevent duplicate or inconsistent customer/product records.
- Handle cancelled orders without incorrectly counting them as completed revenue.
- Validate numeric inputs and payment selections.

Security, Privacy and Ethical Considerations

- Customer records should contain only the information required for the assignment.
- Real payment credentials must never be stored in the R dataset.
- Customer analysis should be used for service improvement rather than unfair discrimination.
- Exports should avoid unnecessary personal information.
- The application should clearly distinguish analytical estimates from business decisions.

Sustainability and Responsible Consumption

- Low-stock and sales analysis can help reduce emergency replenishment and excess inventory.
- Demand information can support more efficient purchasing decisions.
- Digital reporting reduces dependence on repetitive manual calculations.

E. Student Work

1. Problem Understanding and Formulation

1.1 Problem Identification

- The core problem is to connect operational e-commerce transactions with statistical analysis in one R workflow.
- The application must maintain consistent relationships between customers, products and orders while allowing the same records to be analysed for business intelligence.

1.2 Available Information and Data Model

- Customer data contains customer ID, customer name, city and registration date.
- Product data contains product ID, product name, category, price, stock and units sold.
- Order data contains order ID, customer ID, product ID, date, quantity, unit price, payment mode and status.
- Revenue is calculated only from completed orders; cancelled orders are excluded from completed revenue.

1.3 Independent Dataset Used

- The independent demonstration contains 15 customers, 12 products and 36 order records.
- There are 34 completed orders and 2 cancelled orders.
- Completed-order revenue for the demonstration is ₹115,736.
- The mean completed order value is ₹3,404.00.
- The values are synthetic and are created specifically for academic demonstration; they are not real customer transactions.

1.4 Assumptions

- Each order refers to one customer and one product line.
- Unit price is stored with the order so historical order values remain stable if catalogue prices later change.
- A completed order reduces inventory; a valid cancellation reverses the corresponding stock movement.
- The low-stock threshold is 10 units for the prototype.
- Payment mode is stored as a category rather than as sensitive payment credentials.

1.5 Problem Decomposition

1. Customer management
2. Product and inventory management
3. Order processing
4. Discount and return rules
5. Data-frame preparation
6. Sales and customer analysis
7. Visualisation
8. File export and reporting
9. Validation

2. Application of Course Knowledge

2.1 R Data Structures

- Vectors are appropriate for simple fields such as product IDs and prices.
- Lists can represent a flexible object containing multiple related attributes.
- Data frames are used for rectangular customer, product and order datasets.
- Factors can represent categorical fields such as product category, order status and payment method.

```
# Example data-frame construction
products <- data.frame(
  product_id = c("P101", "P102", "P103"),
  product_name = c("AeroBook 14", "Pulse Mouse", "SoundPod Mini"),
  category = c("Electronics", "Electronics", "Electronics"),
  price = c(5899, 799, 1499),
  stock = c(7, 18, 12)
)
```

2.2 Control Structures and Validation

- if/else conditions are used for stock checks, payment validation and discount rules.
- while loops can maintain the menu until the operator chooses Exit.
- for loops can be used for repeated report processing when vectorised alternatives are not convenient.
- Functions isolate validation so the same checks can be reused across menu options.

```
validate_quantity <- function(qty) {
  if (length(qty) != 1 || is.na(qty) || qty <= 0) {
    return(FALSE)
  }
  TRUE
}
```

2.3 Functions and Modular Programming

- register_customer() – creates a validated customer record.
- add_product() – adds a catalogue item.
- place_order() – checks stock and creates an order.
- cancel_order() – changes status and reverses stock when valid.
- calculate_discount() – applies category-specific rules.
- sales_summary() – calculates revenue and order metrics.
- low_stock_report() – identifies products below the threshold.

2.4 Object-Oriented Design in R

- An S3 design can represent Product, Customer and Order objects with class attributes.
- Generic functions can then dispatch category- or object-specific behaviour.

```
new_product <- function(id, name, category, price, stock) {
  x <- list(
    product_id = id,
    product_name = name,
    category = category,
    price = price,
    stock = stock
  )
  class(x) <- "Product"
  x
}
```

```
print.Product <- function(x, ...) {
  cat(x$product_id, "-", x$product_name,
```

```

      "|", x$category, "| Stock:", x$stock, "\n")
}

```

2.5 Category-Specific Business Rules

- Electronics can use a smaller discount but a longer replacement-oriented return window.
- Clothing can use a percentage discount with size/fit-oriented return conditions.
- Grocery products can have shorter return eligibility because of perishability.
- The purpose of the rules is to demonstrate method-based variation, not to define universal retail policy.

```

discount_rate <- function(category, amount) {
  if (category == "Electronics") return(ifelse(amount >= 5000, 0.08, 0.04))
  if (category == "Clothing")    return(ifelse(amount >= 2500, 0.12, 0.06))
  if (category == "Grocery")     return(0.03)
  0.05
}

```

2.6 Data-Frame Operations

- Filtering selects completed orders or low-stock products.
- Sorting ranks customers and products.
- Merging joins order records with product and customer information.
- Grouping and aggregation calculate revenue, quantities and customer totals.
- Reshaping can transform monthly/category summaries into report-friendly structures.

```

# Merge order, product and customer information
order_detail <- merge(orders, products,
                     by = "Product_ID")
order_detail <- merge(order_detail, customers,
                     by = "Customer_ID")

completed <- subset(order_detail, Status == "Completed")
completed$Revenue <- completed$Quantity * completed$Unit_Price

```

2.7 Correlation and Statistical Analysis

- Correlation can be used to examine relationships such as order quantity and revenue.
- A positive association is expected when higher quantities are purchased at similar unit prices, although price variation can weaken the relationship.

```

cor(completed$Quantity, completed$Revenue, use = "complete.obs")

```

2.8 Regression Model

- A simple linear model can estimate completed revenue from quantity and unit price.
- The model is used as an analytical demonstration rather than a production demand-forecasting model.

```

fit <- lm(Revenue ~ Quantity + Unit_Price, data = completed)
summary(fit)

# Prediction for a hypothetical order
new_order <- data.frame(Quantity = 3, Unit_Price = 1299)
predict(fit, newdata = new_order)

```

2.9 Decision Tree Concept

- A decision tree can classify an order or product into a business category using variables such as quantity, price, category and stock.
- The tree structure is easy to interpret and can support rule discovery, although it should be validated before operational deployment.

3. Solution / Design / Methodology

3.1 Overall Architecture

10. Input / menu layer
11. Customer and product storage
12. Order transaction layer
13. Validation and business rules
14. Data-frame transformation
15. Statistical analysis
16. Visualisation
17. CSV report export

3.2 Order Processing Logic

18. Read customer ID, product ID and quantity.
19. Validate that the customer and product exist.
20. Check that the quantity is positive.
21. Check available stock.
22. Determine category and applicable discount.
23. Calculate gross value, discount and net value.
24. Record payment mode and order status.
25. Reduce stock and update units sold.
26. Return a transaction confirmation.

3.3 Cancellation Logic

27. Find the order by order ID.
28. Check that it is currently completed and eligible for cancellation.
29. Change status to Cancelled.
30. Restore the sold quantity to inventory.
31. Exclude the cancelled order from completed-revenue reports.

3.4 Report Design

- Dashboard metrics: completed revenue, completed orders, registered customers and low-stock product count.
- Category report: revenue contribution by category.
- Product report: units sold and stock remaining.
- Customer report: total spending and order frequency.
- Order report: transaction-level detail with status and payment mode.

3.5 Menu Structure

1. Customer Management
2. Product Management
3. Place Order
4. Cancel Order
5. Inventory Status
6. Sales Analytics
7. Customer Analytics
8. Export Reports
9. Exit

4. Use of Modern Tools

4.1 R Tools and Packages

- R – core programming, data structures, functions and statistical modelling.
- RStudio – script development, console execution and project organisation.
- ggplot2 – publication-quality analytical visualisations.
- dplyr – filtering, grouping, summarising and joining data.
- tidyr – reshaping and tidying analytical datasets.
- readr – CSV import and export.
- rpart / rpart.plot – decision-tree modelling and visualisation where required.

4.2 Evidence of Tool Usage

- The code blocks in this assignment are written in R and correspond to the independent dataset and workflow described above.
- The graphs below represent the analytical outputs expected from the same calculations in an R implementation.
- The external CartIQ reference was used for workflow inspiration: dashboard metrics, orders, products, customers, analytics and report-export concepts. The data, wording and visual outputs in this assignment are independently constructed.

Reference Application Observations

The supplied CartIQ application presents a compact e-commerce control-room workflow with a dashboard, order operations, product and inventory monitoring, customer intelligence, analytics and report export. Its dashboard focuses on completed revenue, order count, registered customers and low-stock items.

The Orders module supports placing, inspecting and cancelling transactions, while the Products and Customers modules organise catalogue, stock, purchase frequency and spending information. The Analytics and Reports sections provide category performance, top-customer insights and downloadable order, product, customer and full analytics outputs.

These workflow ideas were used only as functional reference points. The dataset, calculations, wording, charts and implementation presented in this assignment remain independently constructed.

4.3 File I/O

```
write.csv(customers, "customers.csv", row.names = FALSE)
write.csv(products, "products.csv", row.names = FALSE)
write.csv(orders, "orders.csv", row.names = FALSE)
```

```
customers_in <- read.csv("customers.csv")
products_in <- read.csv("products.csv")
orders_in <- read.csv("orders.csv")
```

4.4 Visualisation Code

```
library(ggplot2)
library(dplyr)

monthly <- completed %>%
  mutate(Month = format(as.Date(Order_Date), "%b")) %>%
  group_by(Month) %>%
  summarise(Revenue = sum(Revenue))

ggplot(monthly, aes(Month, Revenue)) +
  geom_col() +
  labs(title = "Monthly Completed Revenue",
```

```
x = "Month", y = "Revenue (₹)" +  
theme_minimal()
```

5. Results and Validation

5.1 Overall Transaction Results

Metric	Independent Result
Registered customers	15
Products in catalogue	12
Total order records	36
Completed orders	34
Cancelled orders	2
Completed revenue	₹115,736
Average completed order	₹3,404.00
Low-stock products (<10 units)	6

5.2 Monthly Revenue

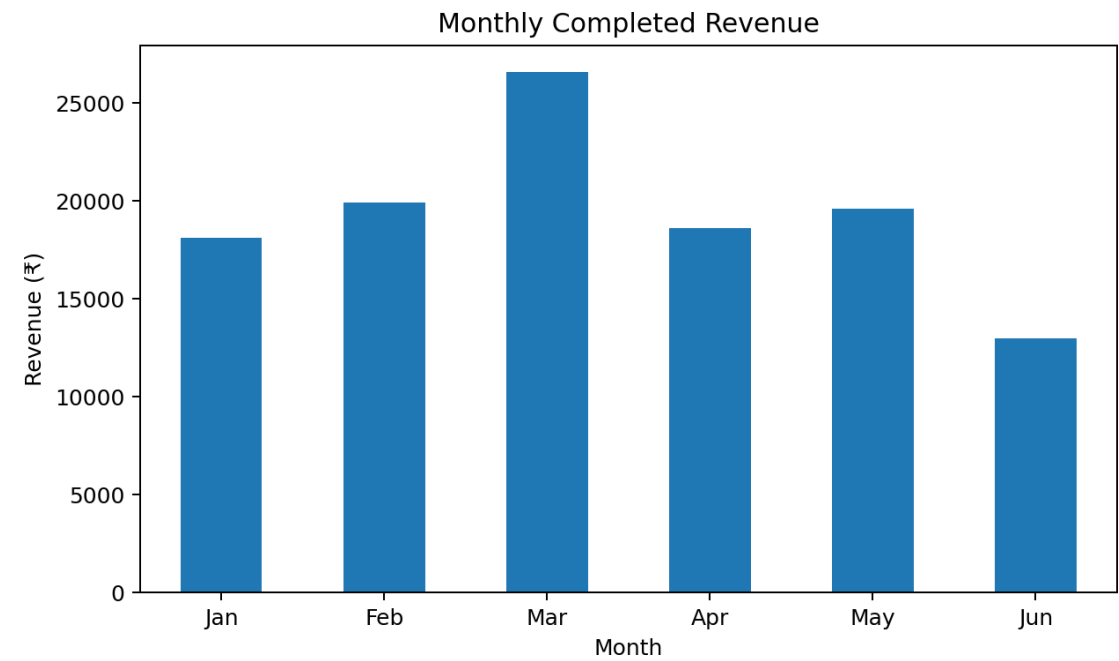


Figure 5.1 — Independent monthly completed-revenue analysis.

5.3 Category Performance

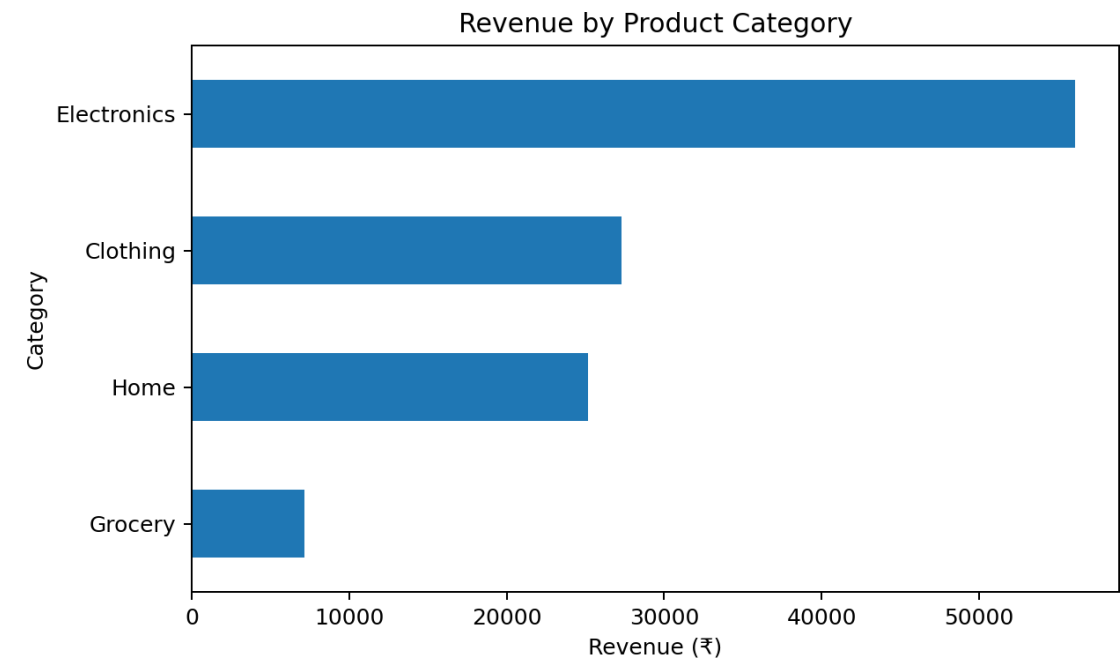


Figure 5.2 — Independent revenue distribution across product categories.

5.4 Best-Selling Products

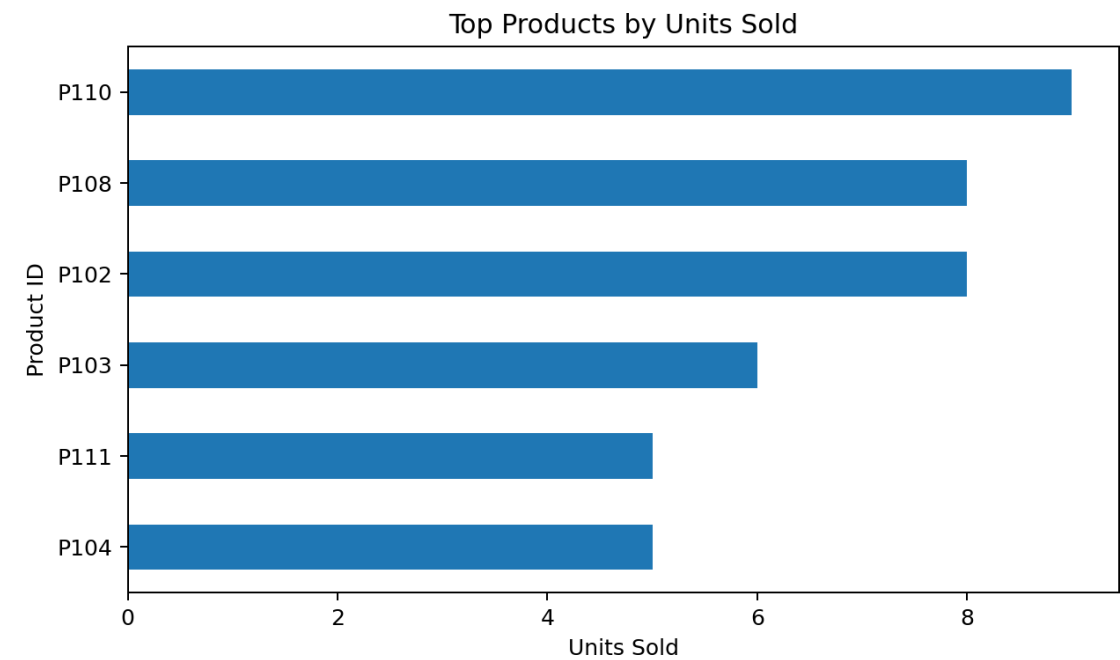


Figure 5.3 — Independent product ranking by units sold.

5.5 Low-Stock Analysis

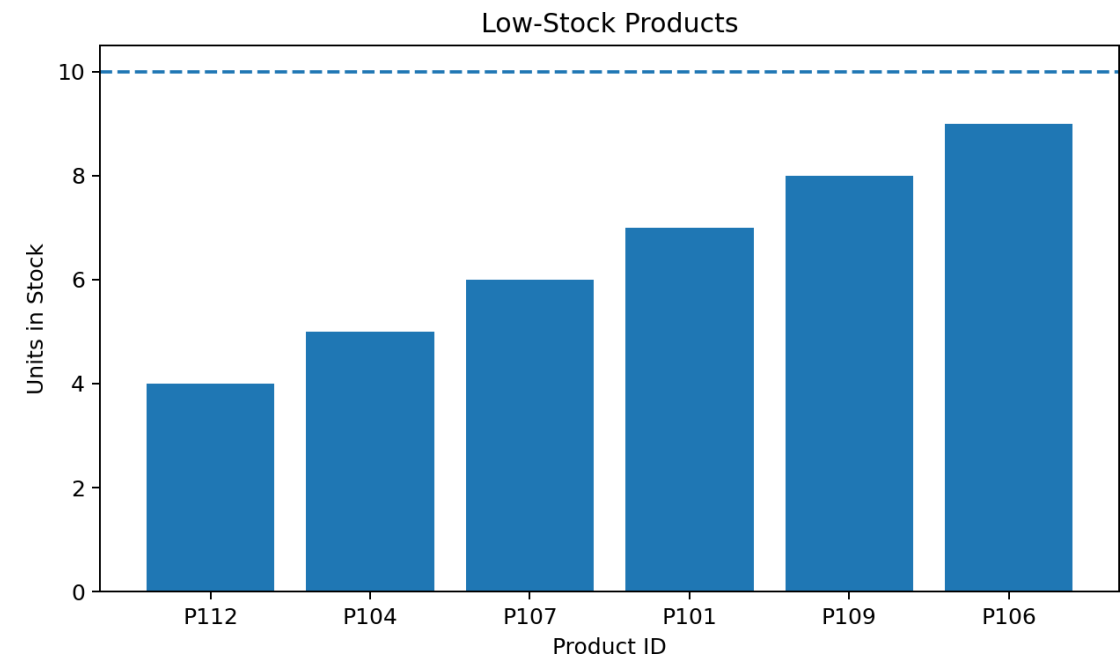


Figure 5.4 — Independent low-stock monitoring output.

5.6 Quantity and Revenue Relationship

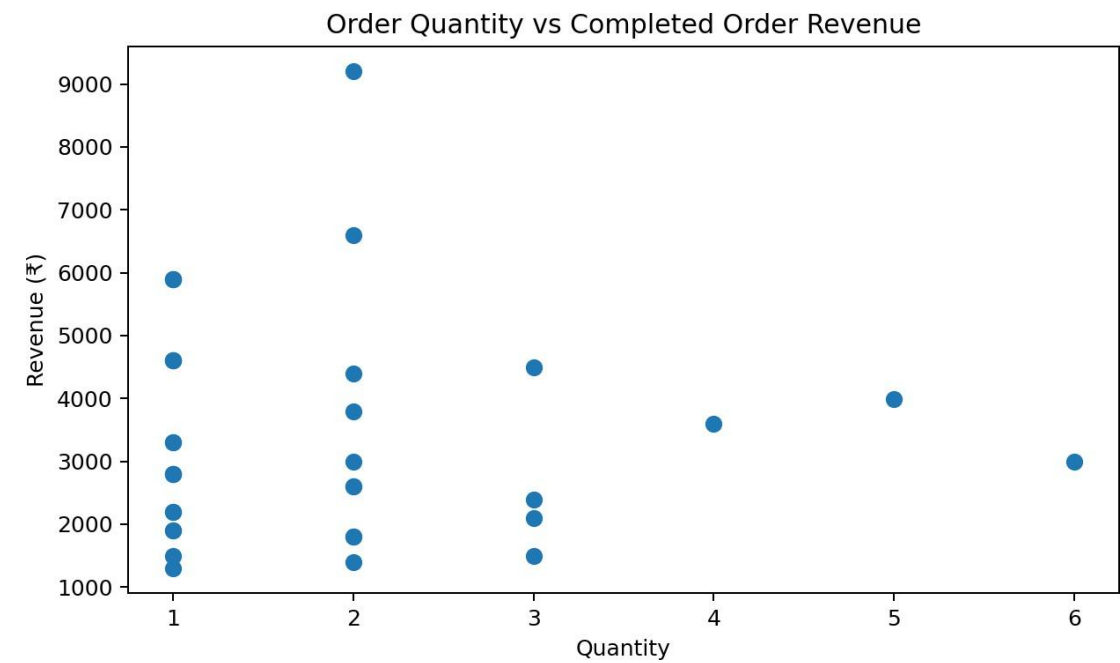


Figure 5.5 — Independent relationship between order quantity and completed revenue.

5.7 Validation Checks

- Cancelled orders are excluded from completed revenue.
- Every completed order references a customer and product present in the respective master data.
- Revenue is calculated as quantity multiplied by unit price before discount adjustments where applicable.
- Low-stock reporting consistently uses the defined threshold of 10 units.

- Customer totals are calculated from completed orders only.
- Product rankings use completed quantities rather than cancelled quantities.
- CSV exports preserve the analytical fields required for later reloading.

5.8 Sample Customer Ranking

Customer_ID	Total_Spending	Completed_Orders
C014	15495	3
C004	12497	2
C002	10696	3
C013	10297	2
C003	10094	3
C009	9390	3
C001	8492	2
C011	8295	3

5.9 Low-Stock Table

Product_ID	Product_Name	Category	Stock
P112	SmartLamp Go	Home	4
P104	ViewMax Monitor 24	Electronics	5
P107	TrailPro Backpack	Clothing	6
P101	AeroBook 14 Laptop	Electronics	7
P109	NutriBlend Mixer	Home	8
P106	Stride Running Shoes	Clothing	9

6. Analysis and Engineering Decision

6.1 Interpretation of Results

- The independent dataset produces ₹115,736 in completed revenue from 34 completed transactions.
- The category plot shows which product groups contribute most to completed revenue, allowing management to prioritise category-level inventory and promotions.
- The product ranking identifies items with higher unit movement, while the low-stock chart highlights products that may require replenishment.
- Customer ranking converts transaction-level records into customer-value information that can support retention and service planning.

6.2 Data-Frame Approach versus Separate Objects

- Data frames are selected for analytical reporting because filtering, joining and aggregation are straightforward.
- S3 objects are used for domain behaviour because they provide a clear way to package customer/product/order attributes and dispatch methods.
- The hybrid approach is preferable to forcing every operation into a single representation.

6.3 Base R versus Tidyverse

- Base R is sufficient for core data structures, functions, control flow and statistical models.
- dplyr and tidyr make repeated grouping, joining and reshaping easier to read.
- ggplot2 provides consistent analytical graphics and improves report presentation.
- The final implementation can use a mixed approach as long as the workflow remains reproducible.

6.4 Advantages

- Modular functions reduce code duplication.
- Data frames support structured analytics.
- S3 classes demonstrate object-oriented R programming.
- CSV I/O makes the prototype portable.
- Visual reports make business patterns easier to interpret.
- Validation reduces the risk of invalid transaction states.

6.5 Limitations

- The dataset is synthetic and relatively small.
- The prototype does not connect to a live payment gateway or production database.
- Discount and return policies are illustrative rather than universal retail policies.
- The regression and decision-tree components are analytical demonstrations and require larger datasets for reliable forecasting.
- A production application would require authentication, database transactions, audit logs and stronger security controls.

6.6 Engineering Decision

- The selected design is a menu-driven R application backed by structured data frames, S3 objects, reusable functions and CSV persistence.
- For analytics, a tidy data workflow with grouped summaries and ggplot2 visualisations is preferred because it keeps the transformation steps readable and reproducible.

- This architecture directly addresses the assignment's requirements while leaving a clear path toward database integration and a graphical interface.

7. Broader Considerations

7.1 Sustainability

- Inventory analytics can reduce unnecessary overstocking and support better replenishment decisions.
- Digital reports reduce repetitive manual calculations and paper-based reporting.

7.2 Society and Customers

- Customer analytics can improve product availability and service quality.
- Customer-level insights should not be used to unfairly exclude or disadvantage individuals.

7.3 Safety and Reliability

- Incorrect stock updates can cause overselling, so order and cancellation logic must maintain consistent inventory.
- Input validation and transaction-status checks are essential for reliable operation.

7.4 Ethics and Privacy

- Only minimum customer information should be retained for analytics.
- Payment credentials must never be stored in the assignment dataset.
- Customer segmentation should be monitored for unfair or discriminatory outcomes.

7.5 Economics

- Revenue, best-seller and low-stock reports support purchasing and promotion decisions.
- Automating repetitive calculations can reduce operational effort.

7.6 Accessibility and Professional Responsibility

- Reports should use readable labels, clear tables and understandable charts.
- Developers are responsible for documenting assumptions and limitations rather than presenting synthetic analysis as real business evidence.

8. Conclusion

- A complete e-commerce analytics problem can be addressed by combining R programming fundamentals with structured data management and statistical analysis.
- The independent prototype models customers, products and orders, supports transaction logic, calculates business metrics and generates analytical visualisations.
- The results demonstrate how R can move from raw transaction records to revenue, product, customer and inventory insights.
- The design satisfies the major functional and analytical requirements of the assignment while remaining suitable for future database and interface extensions.
- The main limitation is that the demonstration uses synthetic data; real deployment would require larger datasets, security controls, testing and business validation.

9. Student Reflection

9.1 Learning Beyond Direct Classroom Theory

- I learned that R programming concepts become more useful when they are connected to a complete application workflow rather than used as isolated examples.
- I understood the importance of separating operational data from analytical summaries so that reports can be recalculated from the source records.
- I learned how functions and S3 classes can reduce repetition and make business rules easier to maintain.
- The visualisation stage showed how statistical results can be converted into practical inventory and sales decisions.

9.2 Challenges Faced

- Maintaining consistent relationships between customer, product and order records.
- Handling completed and cancelled orders correctly in revenue calculations.
- Designing reusable functions without duplicating validation logic.
- Selecting charts that communicate business information without unnecessary complexity.

9.3 Skills Developed

- R data structures and data-frame manipulation.
- Functions, control structures and validation.
- S3 object-oriented programming.
- Data merging, grouping and aggregation.
- CSV file handling.
- Statistical analysis and correlation.
- Regression and decision-tree concepts.
- Business-oriented data visualisation.

9.4 Improvements With Additional Time and Resources

- Connect the R application to SQLite or PostgreSQL for persistent multi-user data storage.
- Add authentication and role-based access for administrators and staff.
- Build a Shiny dashboard with live KPIs, filters and downloadable reports.
- Add time-series forecasting for product demand and replenishment planning.
- Introduce automated unit tests for order, cancellation and inventory functions.
- Use a larger real or properly licensed anonymised dataset for stronger statistical validation.

10. References

32. CartIQ – E-Commerce Intelligence. Supplied reference web application: <https://hk8387121-cpu.github.io/R-Programming-Assignment/>. Dashboard, order, product, customer, analytics and report workflow reference. Accessed 02 September 2026.
33. R Core Team. R version 4.5.1 (Great Square Root), released 13 June 2025. R Foundation for Statistical Computing. Accessed 02 September 2026.
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37. Wickham, H., Chang, W., Henry, L., Pedersen, T. L., Takahashi, K., Wilke, C., Woo, K., Yutani, H., Dunnington, D., and van den Brand, T. (2026). ggplot2 package documentation. tidyverse. Accessed 02 September 2026.
38. ITA0401 – Statistics with R Programming, Assignment 1: E-Commerce Order and Customer Analytics System. Course assignment document, 2026.

Appendix A – Independent Sample Data

- The following sample records are included to make the analysis reproducible and clearly distinguish this version from the supplied reference application.

Order_ID	Customer_ID	Product_ID	Order_Date	Quantity	Unit_Price	Payment	Status
1001	C014	P103	2026-01-04	2	1499	UPI	Completed
1002	C006	P107	2026-01-07	1	3299	Card	Completed
1003	C011	P102	2026-01-10	3	799	UPI	Completed
1004	C002	P109	2026-01-13	1	2199	Cash	Completed
1005	C009	P105	2026-01-18	2	1299	Card	Completed
1006	C003	P104	2026-01-22	1	4599	UPI	Completed
1007	C015	P110	2026-01-28	4	499	Card	Cancelled
1008	C004	P101	2026-02-02	1	5899	UPI	Completed
1009	C012	P108	2026-02-06	2	899	Cash	Completed
1010	C001	P103	2026-02-11	3	1499	Card	Completed
1011	C008	P106	2026-02-15	1	1899	UPI	Completed
1012	C010	P111	2026-02-19	2	699	Card	Completed
1013	C005	P107	2026-02-23	1	3299	UPI	Cancelled
1014	C013	P109	2026-02-27	2	2199	Card	Completed
1015	C007	P112	2026-03-03	1	2799	UPI	Completed
1016	C014	P104	2026-03-08	2	4599	Card	Completed
1017	C002	P101	2026-03-12	1	5899	UPI	Completed
1018	C009	P106	2026-03-17	2	1899	Cash	Completed

Appendix B – Product Master Sample

Product_ID	Product_Name	Category	Price	Stock	Units_Sold
P101	AeroBook 14 Laptop	Electronics	5899	7	21
P102	Pulse Wireless Mouse	Electronics	799	18	37
P103	SoundPod Mini	Electronics	1499	12	42
P104	ViewMax Monitor 24	Electronics	4599	5	19
P105	UrbanFlex Hoodie	Clothing	1299	24	31
P106	Stride Running Shoes	Clothing	1899	9	26
P107	TrailPro Backpack	Clothing	3299	6	17
P108	DailyFresh Grocery Box	Grocery	899	28	54
P109	NutriBlend Mixer	Home	2199	8	23
P110	EcoSip Bottle	Home	499	32	61
P111	KitchenMate Pan	Home	699	14	35
P112	SmartLamp Go	Home	2799	4	14

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

G. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation			L3/L4	10
Application of knowledge			L3/L4	20
Solution / Design / Implementation			L4/L5/L6	20
Modern tool usage			L3/L4	10
Validation			L4/L5	15
Analysis & justification			L4/L5	15
Broader considerations			L4/L5	5
Documentation & reflection			L3/L4	5

Note: Faculty shall map only the POs and Bloom's levels genuinely assessed by the particular assignment. Every assignment is **not required to map to every PO**.

H. Faculty Evaluation & Continuous Improvement

CO Attainment

Performance Level	Criterion
Level 3	$\geq 70\%$
Level 2	60–69%
Level 1	50–59%
Level 0	$< 50\%$

Target CO Attainment: _____

Actual CO Attainment: _____

Faculty Analysis

Areas in which students performed well:

Areas requiring improvement:

Corrective / Improvement Action:

Action to be implemented in the next offering:

Documents to be Maintained

The department/course faculty shall maintain:

1. Assignment question/problem statement
2. CO–PO and Bloom's mapping
3. Assessment rubric
4. Student submissions
5. Tool/simulation/experimental evidence, where applicable
6. Evaluated scripts/reports
7. Marks and rubric-wise assessment
8. CO attainment analysis
9. Sample student work representing different performance levels
10. Faculty analysis and continuous-improvement action